

PROJECT PERFORMANCE TESTING

Date	06 july 2026
Team ID	Individual
Project Name	India's Agricultural Crop Production Analysis (1997-2021) using Tableau
Maximum Marks	4

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	India's Agricultural Crop Production dataset (1997-2021, 3,45,407 records covering 36 states/UTs, 729 districts and 57 crops) with fields State, District, Crop, Year, Season, Area, Area Units, Production, Production Units and Yield, was connected to Tableau via a CSV live connection. Area, Production and Yield rendered as numeric measures; State, District, Crop, Year and Season rendered as dimensions.
2.	Data Preprocessing	No null values or duplicate rows were found. The duplicated "Production Units" header was renamed to "Yield" for clarity. The Year field (e.g. "2001-02") was split into a Start Year integer to allow proper chronological sorting and trend analysis. Area, Production and Yield were confirmed as numeric measures, and District/State names were checked for consistent casing.
3.	Utilization of Filters	Quick filters were applied for State, Crop, Season and Start Year (with a year-range slider covering 1997-2021), along with a parameter to toggle the analysis measure between Area, Production and Yield. Filter actions were configured between dashboard sheets so selecting a state or crop filters the related visualizations.
4.	Calculation fields Used	Calculated fields created: Start Year (parsed from the Year string for sorting/trend lines), Yield per Hectare ($\text{SUM(Production)/SUM(Area)}$), Production Share % by State ($\text{SUM(Production) / TOTAL(SUM(Production))}$), Top 10 Crops by Production (rank-based set), and Year-over-Year Production Growth %.

5.	Dashboard design	No of Visualizations / Graphs - 5 (Line chart of total Production trend from 1997-2021, map of Production by State, bar chart of Top 10 Crops by Production, stacked bar of Area by Season, and KPI cards for total Production, total Area and average Yield per Hectare), arranged with quick filters and filter actions for interactive exploration.
6	Story Design	No of Visualizations / Graphs - 4 (Story points covering dataset overview across 36 states and 57 crops, national production trend over 1997-2021, state-wise and crop-wise leaders in production, and a concluding summary of yield efficiency findings) built from the dashboard sheets to narrate the analysis.